

Bagged Random Subset Methods for Forecast Combination: Theory, Monte Carlo, and Empirical Evidence

Exact Results, Power-Law Excess Risk, and the Cube-Root Law

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Abstract

We study Random Subset Methods (RSM) for forecast combination in a panel of m individual forecasters. The paper proceeds inductively. We first establish exact population results under the common-loading covariance model: a *conservation law* governing how precision-weighted RSM weights redistribute across subsets, an exact formula for the infinite-bagged excess risk $R_k^{\text{bag},\infty}$, monotone decrease of $R_k^{\text{bag},\infty}$ in k (proved under a stated contraction condition, verified numerically), and a first-order condition for the optimal subset size k^* . We then show that the average-subset risk is tractable via Taylor expansion, yielding a square-root optimal subset size $k_{\text{sub}}^* \propto T^{1/2}$. The infinite-bagged risk resists the same treatment because the key conditional expectation has no closed form without further restrictions. We use Monte Carlo experiments with known Σ_e to characterise the functional form of $R_k^{\text{bag},\infty}$ empirically: the power-law family $R_k^{\text{bag},\infty} \approx C z_k^\alpha$, $z_k = 1/k - 1/m$, fits with $R^2 > 0.97$ in all designs. The exponent $\hat{\alpha} \in (1, 2)$ is determined by the heterogeneity ratio H of idiosyncratic variances and implies an optimal subset size $k^* \propto T^{1/(\alpha+1)}$ scaling between $T^{1/3}$ (small heterogeneity, $\alpha \rightarrow 2$) and $T^{1/2}$ (dominant forecaster, $\alpha \rightarrow 1$). We show that this window is not merely a consequence of the power-law fit: an exact two-sided bracket on the excess risk, combined with a marginal-envelope regularity condition, rigorously confines the true optimal subset size to $T^{1/3} \lesssim k^* \lesssim T^{1/2}$, with no parametric assumption on the shape of $R_k^{\text{bag},\infty}$ required. Applying RSM to FRED-MD GDP forecasts yields statistically significant RMSFE reductions of 10–30% over the equal-weighted mean. The Survey of Professional Forecasters—a near-homogeneous panel—confirms that RSM offers little advantage when forecasters are nearly identical in quality.

Contents

1	Scope and Main Message	4
1.1	Motivation and Relation to the Literature	4
2	Forecast-Combination Setup	8
3	Benchmarks, and Why Both Are Special Cases of RSM	8
3.1	Equal-Weighted Mean	8
3.2	Full Optimal Weights	9
3.3	RSM Unifies Both Benchmarks	9
4	RSM: Definition and the Exact Bagging Bridge	9
5	Common-Loading Structure: Exact Theory	10
5.1	Model and Notation	10
5.2	Infinite-Bagged Weights and the Conservation Law	11
5.3	Exact Formula for the Excess Risk	12
5.4	Monotonicity under a Contraction Condition	13
5.5	Discrete Optimality Condition and Continuous Approximation	14
5.6	Alternative Inflation Factor: What Changes?	14
6	What We Can Derive: Tractable Cases	17
6.1	Average-Subset Optimum: Square-Root Rule	17
6.2	Why the Infinite-Bagged Optimum Resists a Parallel Derivation	18
7	Bounding $Q_k^{\text{bag},\infty}$: Taylor Bounds and the Rate Window	18
7.1	The Bracket	19
7.2	Leading-Order Envelope and the Rate Window	20
7.3	Optimal k : Dependence on T , m , and the Precision Profile	20
7.4	Marginal-Envelope Assumption and the Rate Window for k^*	21
8	Monte Carlo Evidence on $Q^{\text{bag},\infty}$: Exogenous Case	23
8.1	Why $R_k^{\text{bag},\infty} \approx Cz_k^\alpha$: Motivation from Exact Properties	23
8.2	How $\hat{\alpha}$ Is Estimated and What It Measures	24
8.3	The Relationship Between Heterogeneity and $\hat{\alpha}$	24
8.4	Optimal k Under the Power-Law Assumption	27
9	Empirical Application: SPF (Near-Homogeneous Case)	29
9.1	Motivation and Design	29
9.2	Results	30

10 Empirical Application: GDP Forecasting with FRED-MD	30
10.1 Data and Setup	30
10.2 Results	30
11 Conclusion	31
A Proofs of Supporting Results	32
A.1 Derivation of $\bar{v}_{k,i}$	32
A.2 Variance of A_S under Uniform Subsampling	32

1 Scope and Main Message

The central question is: in a large panel of m individual forecasters, how many should a practitioner randomly subsample, and how should subsample forecasts be combined, to minimise out-of-sample loss?

Two distinct RSM objects have been studied in the literature. The *average-subset estimator* picks a random size- k subset, fits constrained OLS within it, and reports that single forecast. The *bagged estimator* averages G such draws; as $G \rightarrow \infty$ it converges to the infinite-bagged estimator. Since $w \mapsto w' \Sigma_e w$ is convex, Jensen's inequality gives $Q_k^{\text{bag}, \infty} \leq Q_k^{\text{sub}}$, so bagging always weakly reduces population risk. Yet prior work has largely analysed Q_k^{sub} and assumed a functional form for the bagged excess risk. We derive what can be derived exactly, then use Monte Carlo to characterise what cannot.

Logic of the paper. Sections 2–3 establish the framework and show that the equal-weighted mean and full estimated optimal are special cases of RSM. Section 4 defines RSM and derives the exact finite-bagging bridge. Section 5 exploits the common-loading structure to obtain the conservation law, exact excess-risk formula, monotone decrease of $R_k^{\text{bag}, \infty}$, and a first-order condition for k^* . Section 6 derives the average-subset optimum and shows why the bagged optimum resists a parallel derivation. Section 7 brackets the infinite-bagged excess risk exactly from both sides, derives the implied cube-root and square-root boundary optimisers, and—under an explicit marginal-envelope regularity condition—proves that the true optimal subset size is rigorously confined to this rate window. Section 8 uses Monte Carlo with known Σ_e to motivate the power-law parametrisation of $R_k^{\text{bag}, \infty}$, characterise $\hat{\alpha}$, and validate the implied k^* within that window. Sections 9–10 present the empirical applications.

1.1 Motivation and Relation to the Literature

The forecast combination puzzle and Elliott–Liao (2025). A long-standing puzzle in forecast combination is that the equal-weighted mean of individual forecasts typically outperforms the estimated minimum-variance combination, despite the latter being the solution to a simple constrained OLS problem [??]. ? provide a complementary perspective: rather than attributing the puzzle solely to estimation error, they ask how large the *population* gains from optimal combination can be under empirically motivated restrictions on the forecast-error covariance $\tilde{\Sigma}$. Their model,

$$\tilde{\Omega} = \sigma_\varepsilon^2 \boldsymbol{\mu}' + \tilde{\Sigma}, \tag{1}$$

is precisely our common-loading structure ($q = \sigma_\varepsilon^2$, $D = \tilde{\Sigma}$). Using SPF data to discipline the space of $\tilde{\Sigma}$, they show that for most empirically plausible parameterisations the pop-

ulation gain $Q^{\text{ave}} - Q^{\text{opt}}$ is small relative to estimation error $(m - 1)/T$ — providing a theoretical rationale for the combination puzzle that operates even in population, before any finite-sample noise.

Inflation factors: Elliott–Liao vs RSM. A key ingredient in the puzzle is the *inflation factor* — the multiplier by which the in-sample optimal-combination loss exceeds its true population value because the weights are estimated rather than known. For the full m -weight optimal combination estimated by constrained OLS, the standard degrees-of-freedom correction is

$$I_m(T) = \frac{T - 1}{T - m}, \quad (2)$$

the exact Gaussian correction for estimating $m - 1$ free parameters from T observations. Its leading-order excess over 1 is

$$I_m(T) - 1 \approx \frac{m - 1}{T}, \quad (3)$$

and it is this quantity $(m - 1)/T$ that ? use as their benchmark for the expected estimation cost. They call it the *best case* for OLS because it is the minimum possible estimation error: it assumes Gaussian errors, a correctly specified model, and the full m -dimensional weight vector. Any real-world estimation procedure can only do worse. Their criterion for full optimal combination to be worthwhile is then $Q^{\text{ave}} - Q^{\text{opt}} > (m - 1)/T \cdot (\sigma_\varepsilon^2 + Q^{\text{opt}})$; for most empirically calibrated $\tilde{\Sigma}$ this condition fails, which they show as the rationale for the combination puzzle.

The present paper departs from this framework in a single but critical respect: we use the inflation factor $(T - 1)/(T - k)$ for a *subset* of size $k < m$, not the full-model factor $(T - 1)/(T - m)$. Since $k \ll m$, the RSM inflation is substantially smaller:

$$\frac{T - 1}{T - k} \ll \frac{T - 1}{T - m}, \quad \text{i.e.} \quad \frac{k - 1}{T} \ll \frac{m - 1}{T}. \quad (4)$$

This opens a window of viable gains that the full-model approach cannot access: RSM can produce a net positive out-of-sample improvement even when Elliott–Liao’s benchmark says full OLS combination is not worth estimating, because RSM’s estimation cost is only $(k - 1)/T$ rather than $(m - 1)/T$. Choosing k optimally therefore means choosing the largest subset for which the RSM gain $Q^{\text{ave}} - Q_k^{\text{bag}, \infty}$ exceeds the RSM-specific inflation cost — the central problem this paper solves.

Their Proposition 1 gives a precise characterisation: when $\boldsymbol{\iota}$ is an eigenvector of $\tilde{\Sigma}$ (equivalently, all row sums of $\tilde{\Sigma}$ are equal), the equal-weighted mean *is* the optimal combination and gains are exactly zero. In our notation this corresponds to $V_\delta = 0$, hence $C_\delta = 0$, the near-homogeneous case in which $k^* = 1$ (Remark 9).

Their key positive result (Section 3.2, Proposition 2) concerns the case where gains *are*

large. They characterise the worst-case $\tilde{\Sigma}$ for the equal-weighted mean: the maximiser of $Q^{\text{ave}} - Q^{\text{opt}}$ over the restricted space has a block structure $\Sigma_{11} = I_{m_1}$, $\Sigma_{12} = m_1^{-1} \iota_1 \iota_2'$, and the corresponding optimal weights are

$$w^{\text{opt}} = \begin{pmatrix} (1/m_1) \iota_1 \\ 0 \end{pmatrix}. \quad (5)$$

That is: *take the equal-weighted average of the first m_1 forecasters and discard the remaining $m - m_1$.* This is the Complete Subset estimator of ? (CSR): a fixed subset of size m_1 with equal within-subset weights. Elliott and Liao thus establish that CSR is the *optimal* response when gains from full optimal combination are largest — but they stop short of two important questions: (i) which forecasters should be in the retained subset, and (ii) how large should the subset be?

How RSM improves on the Elliott–Liao prescription. RSM addresses both gaps and goes one step further.

- (i) **Subset selection via randomisation.** Searching deterministically for the best size- m_1 subset is combinatorially intractable ($\binom{m}{m_1}$ candidates). RSM replaces this with uniform random draws and averaging — eliminating subset-selection variance without any search.
- (ii) **Optimal within-subset weights.** Elliott–Liao’s prescription uses equal weights within the retained subset. RSM instead applies constrained minimum-variance OLS within each draw, giving weights $w_{S,i} = \tau_i / A_S$. Since $w_S = \arg \min_{w' \iota = 1} w' \Sigma_S w$, we have $w_S' \Sigma_S w_S \leq (1/k^2) \iota' \Sigma_S \iota$ — optimal weights are always at least as good as equal weights within the same subset. Taking expectations over S : $Q_k^{\text{RSM}} \leq Q_k^{\text{CSR}}$ strictly whenever Σ_e is not a multiple of the identity.
- (iii) **Optimal k^* .** Neither Elliott–Liao nor CSR provide a theory for the optimal subset size. The present paper derives this: $k^* \propto T^{1/(\alpha+1)}$ under the power-law parametrisation of the excess risk, rigorously confined to the window $T^{1/3} \lesssim k^* \lesssim T^{1/2}$ (Theorem 8), with the conservative, parameter-free default $k^* \approx \lfloor \sqrt{T} \rfloor$ at the square-root end of that window for practical use.

Chen and Maung (2023) and the cost of ad hoc k selection. ? develop nonparametric time-varying forecast combination for high-dimensional data. In their empirical application to inflation and unemployment forecasting they include CSR [?] as a benchmark, evaluated at $k \in \{1, 2, 3, 5\}$. Their Table 3 reveals a telling pattern (ASCFE ratios relative to the AR benchmark; below 1 is better):

Series	$k = 1$	$k = 2$	$k = 3$	$k = 5$
Inflation	0.98	0.95	0.93	0.96
Unemployment	1.01	1.02	1.01	0.97

For *unemployment*, performance is monotonically improving through $k = 5$ — the curve is still decreasing at the largest k they consider. For *inflation*, the minimum is near $k = 3$ –4 but was never precisely located; Chen and Maung do not study $k > 5$. In both cases they were on the decreasing side of the U-shaped $\text{RMSFE}(k)$ curve and never found the optimum. The theory in this paper predicts $k^* \approx \lfloor \sqrt{T} \rfloor \approx 10$ –12 at the sample sizes they used ($T \approx 100$ –150), substantially above $k = 5$.

Relation to Bayesian Model Averaging (BMA). Bayesian Model Averaging [??] forms the combined forecast as $\hat{y}^{\text{BMA}} = \sum_j P(M_j|y) \hat{y}_{M_j}$, where models are weighted by their posterior probabilities. In the forecast-combination context, “models” are typically individual forecasters or subsets of forecasters, and posteriors are proportional to the marginal likelihood (often approximated by BIC).

RSM and BMA share the structure of averaging over subsets, but differ in three fundamental ways.

(a) Prior and weighting. BMA assigns posterior-probability weights to subsets, concentrating mass on subsets with high predictive accuracy. RSM uses a *uniform* (flat) prior over all size- k subsets, so every subset receives the same drawing probability $1/\binom{m}{k}$. Under a flat prior over subsets of size k and equal within-subset weights, the BMA average equals the CSR estimator of ?. RSM replaces equal with optimal within-subset weights, strictly improving on this version of BMA.

(b) Within-subset treatment. BMA typically evaluates each model/subset by its marginal likelihood, treating within-subset weights as either equal or MLE-estimated. RSM explicitly minimises within-subset population variance, giving weights $w_{S,i} = \tau_i/A_S$ that are optimal in the mean-squared-error sense. The infinite-bagged RSM weight $\bar{v}_{k,i} = \mathbb{E}_S[\tau_i/A_S \cdot \mathbf{1}_{i \in S}]$ is therefore a *precision-weighted* average of subset memberships, not a likelihood-weighted average of model performances.

(c) Asymptotic behaviour. BMA posteriors concentrate on the “true” model as $T \rightarrow \infty$ (Bernstein–von Mises), becoming degenerate and eventually selecting a single forecaster. RSM with $k \rightarrow m$ converges to the full minimum-variance combination w^{opt} , never collapsing to a single model. RSM’s behaviour as k grows is governed by the excess risk $R_k^{\text{bag}, \infty} \rightarrow 0$, which is fundamentally a frequentist variance-reduction argument rather than a Bayesian consistency result. The optimal k^* derived in this paper is therefore a frequentist bias-variance trade-off, with no natural Bayesian counterpart.

Position of this paper. We formalise RSM as the solution to a well-posed minimum-variance problem, derive exact population results, use Monte Carlo to characterise the functional form of the excess risk, and provide an actionable rule for k^* selection. Relative to Elliott–Liao, we show that RSM with optimal within-subset weights strictly dominates their CSR prescription and derive the missing k^* theory. Relative to Chen–Maung, we provide the theory that explains their empirical U-shape and guides the choice of k . Relative to BMA, RSM is a frequentist minimum-variance estimator with a natural flat-prior interpretation and a tractable optimality theory.

2 Forecast-Combination Setup

Assumption 1 (Forecast model). *There are m individual forecasts $f_t = (f_{1t}, \dots, f_{mt})'$ of a scalar target y_t . Forecast errors $u_{it} = f_{it} - y_t$ satisfy*

$$\mathbb{E}(u_t) = 0, \quad \text{Var}(u_t) = \Sigma_e \succ 0, \quad \mathbb{E}(u_{it}u_{jt'}) = 0 \text{ for } t \neq t'.$$

For any weight vector w with $w'\boldsymbol{\iota} = 1$ where $\boldsymbol{\iota} = (1, \dots, 1)'$, the forecast $w'f_t$ has mean-squared prediction error

$$\text{MSPE}(w) = w'\Sigma_e w; :=; Q(w). \tag{6}$$

The scalar $s = \text{Var}(y_t - \mathbb{E}[y_t|f_t])$ is the irreducible forecast uncertainty shared by all combinations and plays no role in choosing w . We therefore focus on $Q(w)$ throughout.

3 Benchmarks, and Why Both Are Special Cases of RSM

3.1 Equal-Weighted Mean

The equal-weighted combination assigns weight $1/m$ to every forecaster:

$$w^{\text{ave}} = \frac{1}{m}\boldsymbol{\iota}, \quad Q^{\text{ave}} = \frac{1}{m^2}\boldsymbol{\iota}'\Sigma_e\boldsymbol{\iota}. \tag{7}$$

Since no parameters are estimated, Q^{ave} is the population MSPE with no finite-sample inflation.

3.2 Full Optimal Weights

The infeasible minimum-variance combination is

$$w^{\text{opt}} = \frac{\Sigma_e^{-1}\boldsymbol{\iota}}{\boldsymbol{\iota}'\Sigma_e^{-1}\boldsymbol{\iota}}, \quad Q^{\text{opt}} = (\boldsymbol{\iota}'\Sigma_e^{-1}\boldsymbol{\iota})^{-1}. \quad (8)$$

When Σ_e is estimated from T observations, the degrees-of-freedom inflation of the constrained OLS estimator gives

$$L^{\text{full}} \approx (s + Q^{\text{opt}}) \frac{T-1}{T-m}, \quad (9)$$

which diverges as $m \rightarrow T$, reflecting the curse of dimensionality.

3.3 RSM Unifies Both Benchmarks

We defer the formal definition to Section 4, but note immediately the two limiting cases that motivate RSM as the natural unifying object:

- **Full pool** ($k = m$): the only subset of size m is the full panel, so the RSM weight equals w^{opt} and $Q_m^{\text{bag},\infty} = Q^{\text{opt}}$.
- **Homogeneous forecasters** ($\Sigma_e = \sigma^2[q\boldsymbol{\iota}\boldsymbol{\iota}' + I_m]$ up to a scale): within any subset the constrained OLS weights are equal, so the bagged weight converges to w^{ave} and $Q_k^{\text{bag},\infty} = Q^{\text{ave}}$ for all k .

For $1 \leq k < m$ and non-trivial heterogeneity, RSM interpolates between the two, strictly improving on Q^{ave} while avoiding the curse of dimensionality that makes L^{full} large when m is close to T .

4 RSM: Definition and the Exact Bagging Bridge

Definition 1 (RSM). Let $S \subset \{1, \dots, m\}$ be a random subset of size k , drawn uniformly without replacement from $\binom{m}{k}$ possible subsets. Let $\Sigma_{e,S}$ denote the $k \times k$ principal submatrix of Σ_e indexed by S . The *RSM weight vector for subset S* , embedded in $\mathbb{R}^m$, is

$$v_S = P_S \frac{\Sigma_{e,S}^{-1}\boldsymbol{\iota}}{\boldsymbol{\iota}'\Sigma_{e,S}^{-1}\boldsymbol{\iota}}, \quad (10)$$

where P_S places the k computed weights in their original coordinates and sets the remaining $m - k$ coordinates to zero. The *average-subset* forecast uses a single draw S . The *finite-bagged* forecast with G draws uses $\hat{v}_{k,G} = G^{-1} \sum_{g=1}^G v_{S_g}$, and the *infinite-bagged* weight is

$$\bar{v}_k = \mathbb{E}_S[v_S], \quad (11)$$

where the expectation is over the uniform distribution on subsets of size k .

The three population risks are

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Sigma_e v_S], \quad (12)$$

$$Q_k^{\text{bag}, G} = \mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}], \quad (13)$$

$$Q_k^{\text{bag}, \infty} = \bar{v}_k' \Sigma_e \bar{v}_k. \quad (14)$$

Proposition 1 (Exact finite-bagging bridge). *For G independent draws $S_1, \dots, S_G$,*

$$Q_k^{\text{bag}, G} = Q_k^{\text{bag}, \infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty}), \quad (15)$$

and hence $Q_k^{\text{bag}, \infty} \leq Q_k^{\text{bag}, G} \leq Q_k^{\text{sub}}$ for all $G \geq 1$.

Proof. Let $V = v_S$ and $\bar{v} = \mathbb{E}[V]$. For G i.i.d. draws, $\hat{v}_{k,G} = G^{-1} \sum_g V_g$. Because $\Sigma_e \succ 0$,

$$\begin{aligned} \mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}] &= \bar{v}' \Sigma_e \bar{v} + \mathbb{E}[(\hat{v}_{k,G} - \bar{v})' \Sigma_e (\hat{v}_{k,G} - \bar{v})] \\ &= Q_k^{\text{bag}, \infty} + \frac{1}{G} \mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})]. \end{aligned}$$

Expanding $\mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})] = \mathbb{E}[V' \Sigma_e V] - \bar{v}' \Sigma_e \bar{v} = Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty}$ gives (15). $\square$

Remark 1. Identity (15) is exact: no distributional assumption on $\{f_t\}$ and no approximation. It shows that increasing G monotonically reduces population risk until the infinite-bagged limit is reached, and that the benefit of bagging equals $G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty})$.

5 Common-Loading Structure: Exact Theory

5.1 Model and Notation

Assumption 2 (Common-loading covariance). *The forecast-error covariance matrix satisfies*

$$\Sigma_e = q \boldsymbol{\mu} \boldsymbol{\mu}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad q \geq 0, \quad \sigma_i^2 > 0. \quad (16)$$

Define precision weights $\tau_i = \sigma_i^{-2}$, $A_S = \sum_{i \in S} \tau_i$, and $A_m = \sum_{i=1}^m \tau_i$.

Under this model, for any subset S of size k , the Woodbury identity yields $\Sigma_{e,S}^{-1} = D_S^{-1} - q(1 + qA_S)^{-1} \boldsymbol{\tau}_S \boldsymbol{\tau}_S'$, where $\boldsymbol{\tau}_S = (\tau_i)_{i \in S}$. The constrained optimal weight within S is

$$w_{S,i} = \frac{\tau_i}{A_S}, \quad i \in S, \quad (17)$$

which does not depend on the common-loading parameter q . Consequently the subset risk is

$$Q(w_S) = q + \frac{1}{A_S}; \quad \text{:=}; \quad Q(S). \quad (18)$$

The oracle risk and equal-weight risk are

$$Q^{\text{opt}} = q + \frac{1}{A_m}, \quad Q^{\text{ave}} = q + \frac{\sum_i \tau_i^{-1}}{m^2}. \quad (19)$$

These expressions can be written more compactly in terms of the two scalar parameters used throughout the Monte Carlo experiments:

$$B := \frac{1}{\bar{\tau}} = \frac{m}{A_m}, \quad H := \overline{\sigma^2} \bar{\tau}, \quad (20)$$

where $\bar{\tau} = A_m/m$ is the mean precision and $\overline{\sigma^2} = m^{-1} \sum_i \sigma_i^2$ is the mean idiosyncratic variance. B is the mean idiosyncratic variance (the common scale of the σ_i^2) and $H \geq 1$ measures how spread out the σ_i^2 are ($H = 1$ iff all σ_i^2 are equal). Substituting $\sum_i \tau_i^{-1} = m \overline{\sigma^2} = mHB$ gives

$$Q^{\text{opt}} = q + \frac{B}{m}, \quad Q^{\text{ave}} = q + \frac{HB}{m}, \quad Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m}. \quad (21)$$

The inequality $H \geq 1$ holds because the AM–HM inequality applied to $\{\sigma_i^2\}$ gives $\overline{\sigma^2} \geq m / \sum_i \tau_i = 1/\bar{\tau}$, so $H = \overline{\sigma^2} \bar{\tau} \geq 1$, with equality iff all σ_i^2 are equal. Hence $Q^{\text{ave}} \geq Q^{\text{opt}}$ always, with equality only in the homogeneous case.

5.2 Infinite-Bagged Weights and the Conservation Law

For $i = 1, \dots, m$, define

$$g_i(k) = k \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right], \quad (22)$$

so that the i -th component of the infinite-bagged weight is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \frac{k}{m} \mathbb{E}_S \left[\frac{\tau_i}{A_S} \mid i \in S \right] = \frac{\tau_i g_i(k)}{m}. \quad (23)$$

Since $\sum_{i \in S} w_{S,i} = 1$ for every S , we have $\sum_i \bar{v}_{k,i} = 1$, and therefore $\sum_i \tau_i g_i(k) = m$ for all k .

Proposition 2 (Conservation law). *For every $k \in \{1, \dots, m\}$,*

$$\sum_{i=1}^m \tau_i \left[g_i(k) - \frac{1}{\tau_i} \right] = 0. \quad (24)$$

That is, the precision-weighted deviations of $g_i(k)$ from $1/\tau_i$ sum to zero across all m forecasters.

Proof. From (23), $\sum_i \tau_i g_i(k) = m$. The oracle weight satisfies $\sum_i \tau_i \cdot (1/\tau_i) = m$. Subtracting gives $\sum_i \tau_i [g_i(k) - 1/\tau_i] = 0$. $\square$

Remark 2. *The conservation law has a clear interpretation. The value $1/\tau_i$ is the g -space reference that delivers equal bagged weights: at $k = 1$, $g_i(1) = 1/\tau_i$ and $\bar{v}_{1,i} = 1/m$ for all i . The deviation $\tau_i[g_i(k) - 1/\tau_i] = m\bar{v}_{k,i} - 1$ therefore measures how much more weight forecaster i receives above the equal-weight benchmark, precision-rescaled. The law says these excess weights are zero-sum: RSM redistributes weight relative to equal weighting without any net precision-weighted gain. Note that this differs from the oracle deviation, which uses $g_i^{\text{opt}} = 1/\bar{\tau}$ (uniform across i) rather than $1/\tau_i$ (heterogeneous); the excess-risk formula (26) measures those oracle deviations. It holds for all k and all precision profiles $\{\tau_i\}$, with no distributional assumption.*

5.3 Exact Formula for the Excess Risk

Define the weight deviation and the excess risk:

$$\Delta_{i,k} = \bar{v}_{k,i} - w_i^{\text{opt}}, \quad R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}. \quad (25)$$

Proposition 3 (Exact excess risk). *Under Assumption 2,*

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2, \quad (26)$$

where $\bar{\tau} = A_m/m$ is the mean precision.

Proof. We have $R_k^{\text{bag},\infty} = (\bar{v}_k - w^{\text{opt}})' \Sigma_e (\bar{v}_k - w^{\text{opt}})$. Under the common-loading structure, $\Sigma_e = q\boldsymbol{\iota}\boldsymbol{\iota}' + D$, so

$$R_k^{\text{bag},\infty} = q(\boldsymbol{\iota}'\Delta_k)^2 + \Delta_k' D \Delta_k,$$

where $\Delta_k = \bar{v}_k - w^{\text{opt}}$. Since $\sum_i \bar{v}_{k,i} = 1 = \sum_i w_i^{\text{opt}}$, we have $\boldsymbol{\iota}'\Delta_k = 0$, so the common-loading term vanishes. Therefore

$$R_k^{\text{bag},\infty} = \sum_i \sigma_i^2 \Delta_{i,k}^2 = \sum_i \tau_i^{-1} (\bar{v}_{k,i} - w_i^{\text{opt}})^2.$$

Substituting $\bar{v}_{k,i} = \tau_i g_i(k)/m$ and $w_i^{\text{opt}} = \tau_i/A_m = \tau_i/(m\bar{\tau})$:

$$\begin{aligned} \Delta_{i,k} &= \frac{\tau_i}{m} \left(g_i(k) - \frac{1}{\bar{\tau}} \right), \\ R_k^{\text{bag},\infty} &= \sum_i \tau_i^{-1} \cdot \frac{\tau_i^2}{m^2} \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2 = \frac{1}{m^2} \sum_i \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2. \quad \square \end{aligned}$$

Remark 3 (Boundary conditions). *Formula (26) immediately gives both exact boundary conditions:*

- $k = m$: the only subset is the full panel, so $w_{S,i} = \tau_i/A_m = w_i^{\text{opt}}$ and $g_i(m) = 1/\bar{\tau}$ for all i . Hence $R_m^{\text{bag},\infty} = 0$. ✓
- $k = 1$: subset $\{i\}$ occurs with probability $1/m$, so $\bar{v}_{k,i} = 1/m$ and $g_i(1) = m/(m\tau_i) = 1/\tau_i$. Hence $R_1^{\text{bag},\infty} = (m\bar{\tau})^{-2} \sum_i \tau_i (1/\tau_i - 1/\bar{\tau})^2 > 0$ when variances are not all equal. ✓

5.4 Monotonicity under a Contraction Condition

Proposition 4 (Monotonicity). *Under Assumption 2 with at least two distinct τ_i values, if for every i and every $k \in \{1, \dots, m-1\}$:*

- (a) $g_i(k)$ lies strictly between its $k = 1$ value $1/\tau_i$ and $1/\bar{\tau}$ (the contraction condition); and
- (b) $\dot{g}_i(k) \equiv g_i(k+1) - g_i(k)$ moves $g_i(k)$ toward $1/\bar{\tau}$ (so $\dot{g}_i > 0$ when $g_i < 1/\bar{\tau}$ and $\dot{g}_i < 0$ when $g_i > 1/\bar{\tau}$);

then $R_k^{\text{bag},\infty}$ is strictly decreasing in k : $R_{k+1}^{\text{bag},\infty} < R_k^{\text{bag},\infty}$. Both conditions are verified numerically across all simulation designs.

Proof. Let $\dot{g}_i(k) = g_i(k+1) - g_i(k)$. From (23), differencing $\sum_i \tau_i g_i(k) = m$ gives $\sum_i \tau_i \dot{g}_i(k) = 0$. From (26),

$$\begin{aligned} R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} &= \frac{1}{m^2} \sum_i \tau_i \left[\left(g_i(k+1) - \frac{1}{\bar{\tau}} \right)^2 - \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2 \right] \\ &= \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k) \left[2 \left(g_i(k) - \frac{1}{\bar{\tau}} \right) + \dot{g}_i(k) \right]. \end{aligned}$$

We use $\sum_i \tau_i \dot{g}_i(k) = 0$ to subtract any constant from $g_i(k) - 1/\bar{\tau}$, so

$$R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} = \frac{2}{m^2} \sum_i \tau_i \dot{g}_i(k) \left(g_i(k) - \frac{1}{\bar{\tau}} \right) + \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k)^2.$$

We re-factor the two terms together:

$$R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k) \left[\left(g_i(k+1) - \frac{1}{\bar{\tau}} \right) + \left(g_i(k) - \frac{1}{\bar{\tau}} \right) \right].$$

Under conditions (a)–(b), $g_i(k)$ and $g_i(k+1)$ both lie strictly on the same side of $1/\bar{\tau}$, while $\dot{g}_i(k)$ has the opposite sign. Hence every summand $\tau_i \dot{g}_i(k) [(g_i(k+1) - 1/\bar{\tau}) + (g_i(k) - 1/\bar{\tau})] < 0$, and the difference is strictly negative. $\square$

Remark 4. *Proposition 4 relies on structural properties (a)–(b) above, which are argued analytically in limiting cases and confirmed numerically across all simulation designs. A fully algebraic proof of these properties for general precision profiles $\{\tau_i\}$ remains open.*

5.5 Discrete Optimality Condition and Continuous Approximation

With the finite-sample inflation factor $I(T, k) = (T - 1)/(T - k)$ applied to account for estimating k free combination weights from T observations, the total loss is

$$L_k^{\text{bag}, \infty} = (s + Q_k^{\text{bag}, \infty}) \frac{T - 1}{T - k} = (A + R_k^{\text{bag}, \infty}) \frac{T - 1}{T - k}, \quad A = s + Q^{\text{opt}}. \quad (27)$$

Proposition 5 (First-Order Condition). *The integer minimiser k^* of $L_k^{\text{bag}, \infty}$ over $\{1, \dots, m\}$ satisfies, approximately,*

$$(T - k^*) (R_{k^*}^{\text{bag}, \infty} - R_{k^*+1}^{\text{bag}, \infty}) \approx A + R_{k^*}^{\text{bag}, \infty}, \quad A = s + Q^{\text{opt}}. \quad (28)$$

Proof. At the interior optimum k^* , the loss must satisfy $L_{k^*} \leq L_{k^*+1}$ and $L_{k^*} \leq L_{k^*-1}$. The condition $L_{k^*} \leq L_{k^*+1}$ gives

$$(A + R_{k^*}) \frac{T - 1}{T - k^*} \leq (A + R_{k^*+1}) \frac{T - 1}{T - k^* - 1}.$$

Cross-multiplying and rearranging:

$$\begin{aligned} (A + R_{k^*})(T - k^* - 1) &\leq (A + R_{k^*+1})(T - k^*) \\ (R_{k^*} - R_{k^*+1})(T - k^*) &\leq A + R_{k^*}. \end{aligned}$$

Similarly, $L_{k^*} \leq L_{k^*-1}$ gives $(R_{k^*-1} - R_{k^*})(T - k^*) \geq A + R_{k^*}$. These two inequalities bracket the optimum:

$$(T - k^*)(R_{k^*-1} - R_{k^*}) \geq A + R_{k^*} \geq (T - k^*)(R_{k^*} - R_{k^*+1}).$$

Replacing discrete differences by a continuous derivative and setting to equality yields the continuous approximation (28). This requires no parametric assumption on $R_k^{\text{bag}, \infty}$, but is approximate since the integer problem is characterised by inequalities. $\square$

Remark 5. Equation (28) equates the marginal gain from increasing k by 1 (left side: risk reduction $R_{k^*} - R_{k^*+1}$ accrues over the remaining $T - k^*$ periods) to the current total loss level (right side: $A + R_{k^*} = s + Q^{\text{opt}} + R_{k^*}$). It holds for any shape of $R_k^{\text{bag}, \infty}$ and does not require the power-law parametrisation introduced later.

5.6 Alternative Inflation Factor: What Changes?

The inflation factor $I(T, k) = (T - 1)/(T - k)$ is the exact degrees-of-freedom correction for constrained OLS with k free parameters estimated from T observations. Two alternatives appear in the literature and in practice:

Alternative 1: Full-model inflation $I(m, T) = (T - 1)/(T - m)$. Some practitioners apply a single penalty equal to that of the full estimated model, treating every subset as if it uses all m parameters regardless of its actual size k . With this factor:

$$\tilde{L}_k = (A + R_k^{\text{bag}, \infty}) \frac{T - 1}{T - m}, \quad (29)$$

the multiplier $(T - 1)/(T - m)$ is *constant in k* . Therefore $\tilde{L}_k$ is minimised by the same k that minimises $R_k^{\text{bag}, \infty}$, which is $k^* = m$ (since $R_k^{\text{bag}, \infty}$ is strictly decreasing and $R_m^{\text{bag}, \infty} = 0$). The full-model inflation factor thus destroys the interior optimum: it always recommends the full estimated optimal combination, which is the combination most exposed to the curse of dimensionality. The k -dependence of the standard factor is not a modelling convenience but a structural requirement for RSM to have a non-trivial optimum.

Alternative 2: Linear approximation $I_{\text{lin}}(T, k) = 1 + (k - 1)/T$. For $k \ll T$, the first-order Taylor expansion of $(T - 1)/(T - k)$ around $k = 0$ gives

$$\frac{T - 1}{T - k} = \frac{1}{1 - k/T} \cdot \frac{T - 1}{T} \approx \left(1 + \frac{k}{T}\right) \left(1 - \frac{1}{T}\right) \approx 1 + \frac{k - 1}{T} ; :=; I_{\text{lin}}(T, k). \quad (30)$$

Unlike Alternative 1, I_{lin} depends on k and grows with k , so the interior optimum is preserved.

Proposition 6 (Equivalence under linear inflation). *Under the linear approximation $I_{\text{lin}}(T, k) = 1 + (k - 1)/T$:*

(i) *The average-subset optimal subset size satisfies*

$$k_{\text{sub}, \text{lin}}^* = \sqrt{\frac{B(T - 1)}{A_0}} \approx \sqrt{\frac{BT}{A_0}} = k_{\text{sub}}^*,$$

identical to the exact formula (36) to leading order.

(ii) *The power-law optimal subset size satisfies*

$$k_{\text{bag}, \text{lin}}^* \approx \left(\frac{C\alpha T}{A}\right)^{1/(\alpha+1)} = k_{\text{bag}}^*,$$

identical to (55) to leading order.

Quantitative differences appear only at order k/T , which is small when $k \ll T$.

Proof. (i) Under I_{lin} , the average-subset loss is $\tilde{L}_k^{\text{sub}} = (A_0 + B/k)(1 + (k - 1)/T)$. Setting

$\partial \tilde{L}_k / \partial k = 0$:

$$-\frac{B}{k^2} \left(1 + \frac{k-1}{T}\right) + \frac{1}{T} \left(A_0 + \frac{B}{k}\right) = 0.$$

Multiplying through by $k^2 T$ and expanding:

$$-BT - B(k-1) + A_0 k^2 + Bk = 0 \implies A_0 k^2 = B(T-1),$$

giving $k^* = \sqrt{B(T-1)/A_0}$. The exact formula (36) gives $k^* = (-B + \sqrt{B^2 + A_0 B T})/A_0 \approx \sqrt{BT/A_0}$ for large T . Both are $O(T^{1/2})$ and agree to leading order.

(ii) Under I_{lin} , the bagged loss is $(A + C z_k^\alpha)(1 + (k-1)/T)$. For $k \ll T$ the factor $1 + (k-1)/T \approx 1$ is nearly constant, so the FOC at leading order is the same as under the exact factor: $A/T = C\alpha/k^{\alpha+1}$, giving the same k^* . $\square$

When do the two factors differ numerically? The discrepancy between $(T-1)/(T-k)$ and $1 + (k-1)/T$ is

$$\frac{T-1}{T-k} - \left(1 + \frac{k-1}{T}\right) = \frac{(k-1)^2}{T(T-k)}. \quad (31)$$

For $k = 10$, $T = 200$: discrepancy ≈ 0.004 (0.4% of the baseline loss $A \approx 2$). For $k = 30$, $T = 200$: discrepancy ≈ 0.052 (2.6%). The linear approximation is accurate for the exogenous regime ($k^* \approx 2-6$) and remains reasonable at the square-root end of the rate window ($k = \lfloor \sqrt{T} \rfloor \approx 9-14$), but becomes material for substantially larger k (e.g., $k \gtrsim 26-32$), where the exact factor is preferred.

Table 1: Inflation factors and implied k^* under the two approximations. Parameters: $A = 2.06$, $B = 1/\bar{\tau}$, C and α as in the H=6 exogenous DGP.

Inflation factor	Interior optimum?	$k^*(T = 200)$	$k^*(T = 500)$	$k^*(T = 800)$
$(T-1)/(T-k)$ [exact]	Yes	3	5	6
$1 + (k-1)/T$ [linear]	Yes	3	5	6
$(T-1)/(T-m)$ [full]	No	$m = 50$	$m = 50$	$m = 50$
k^* values from grid search over $k = 1, \dots, m$ minimising the respective loss function at the exogenous H=6 DGP parameters. At leading order, exact and linear give identical k^* .				

The main takeaway is clear: the choice between the exact and linear inflation factors has negligible effect on the optimal k^* (both yield the same integer grid minimiser in our designs). What matters is that the factor is *increasing in k* ; any such factor preserves the interior optimum and the T-scaling exponent, while a constant factor destroys both.

6 What We Can Derive: Tractable Cases

6.1 Average-Subset Optimum: Square-Root Rule

From (12) and (18), $Q_k^{\text{sub}} = q + \mathbb{E}_S[1/A_S]$.

Lemma 1 (Taylor expansion of the expected inverse precision sum). *Let $\bar{\tau} = A_m/m$, $\tau_{\min} = \min_i \tau_i$, and $\epsilon_S = A_S - k\bar{\tau}$. Then*

$$\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + R_{3,k}, \quad (32)$$

where $\text{Var}_S(A_S) = k(1 - \frac{k-1}{m-1}) \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau})^2$ and

$$|R_{3,k}| \leq \frac{\mathbb{E}_S|\epsilon_S|^3}{(k\tau_{\min})^4}. \quad (33)$$

Equivalently, if $\tau_{\min}/\bar{\tau} \geq c > 0$ uniformly, then $R_{3,k} = O(\mathbb{E}_S|\epsilon_S|^3/(k\bar{\tau})^4)$.

Proof. Write $A_S = k\bar{\tau} + \epsilon_S$ where $\epsilon_S = A_S - k\bar{\tau}$. Under uniform sampling without replacement, $\mathbb{E}_S[\epsilon_S] = 0$ and $\text{Var}_S(\epsilon_S) = k \cdot \frac{m-k}{m(m-1)} \cdot \sum_i (\tau_i - \bar{\tau})^2 =: \sigma_\epsilon^2$. Taylor's theorem applied to $f(x) = 1/(k\bar{\tau} + x)$ around $x = 0$ gives, for each subset S ,

$$\frac{1}{A_S} = \frac{1}{k\bar{\tau}} - \frac{\epsilon_S}{(k\bar{\tau})^2} + \frac{\epsilon_S^2}{(k\bar{\tau})^3} + r_S,$$

where $r_S = -\epsilon_S^3/(k\bar{\tau} + \theta_S \epsilon_S)^4$ for some $\theta_S \in (0, 1)$. Since $k\bar{\tau} + \theta_S \epsilon_S = (1 - \theta_S)k\bar{\tau} + \theta_S A_S \geq k\tau_{\min}$, $|r_S| \leq |\epsilon_S|^3/(k\tau_{\min})^4$. Taking $\mathbb{E}_S[\cdot]$ and using $\mathbb{E}[\epsilon_S] = 0$, $\mathbb{E}[\epsilon_S^2] = \sigma_\epsilon^2$ gives the stated expansion and the bound (33). The variance formula for hypergeometric sampling completes the proof. $\square$

The leading term $1/(k\bar{\tau})$ dominates when heterogeneity is small. Defining $z_k = 1/k - 1/m$ and noting $1/(k\bar{\tau}) - 1/(m\bar{\tau}) = z_k/\bar{\tau}$:

$$Q_k^{\text{sub}} - Q^{\text{opt}} \approx \frac{z_k}{\bar{\tau}} \propto z_k. \quad (34)$$

The average-subset excess risk decays *linearly* in $z_k \approx 1/k$ for $k \ll m$.

The total average-subset loss with inflation is $L_k^{\text{sub}} \approx (A_0 + B/k)(T-1)/(T-k)$ where $A_0 = s + q$ and $B = 1/\bar{\tau}$. The first-order condition $\partial L_k^{\text{sub}}/\partial k = 0$ gives

$$A_0 k^2 + 2Bk - BT = 0, \quad (35)$$

with positive root

$$k_{\text{sub}}^* = \frac{-B + \sqrt{B^2 + A_0 BT}}{A_0} \sim \left(\frac{BT}{A_0} \right)^{1/2} \propto T^{1/2}. \quad (36)$$

The average-subset estimator has a square-root optimal subset size. This leading-order result requires only the common-loading structure; higher Taylor terms introduce a variance correction of relative order $(\sigma_\epsilon/k\bar{\tau})^2$, with the remaining error controlled by (33); these terms are negligible when precision heterogeneity is moderate and the precision profile is bounded away from zero.

6.2 Why the Infinite-Bagged Optimum Resists a Parallel Derivation

For the infinite-bagged estimator, the excess risk is $R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i (g_i(k) - 1/\bar{\tau})^2$, where

$$g_i(k) = k \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right]. \quad (37)$$

The average-subset derivation above used $\mathbb{E}_S[1/A_S]$, an unconditional expectation that factored cleanly. Here we need $\mathbb{E}_S[1/A_S \mid i \in S]$, the conditional expectation of $1/A_S$ given that forecaster i is present.

When $i \in S$, the sum $A_S = \tau_i + A_{S \setminus \{i\}}$, where $A_{S \setminus \{i\}} = \sum_{j \in S, j \neq i} \tau_j$ is the sum of $k-1$ other precisions drawn from $\{\tau_j\}_{j \neq i}$. Taylor-expanding $1/A_S$ around $A_{S \setminus \{i\}}$:

$$\mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right] = \mathbb{E} \left[\frac{1}{\tau_i + A_{S \setminus \{i\}}} \right], \quad (38)$$

which is a harmonic mean shifted by τ_i . This depends on the entire distribution of precision sums over size- $(k-1)$ subsets of $\{\tau_j\}_{j \neq i}$ — a different distribution for each i . There is no general closed form for this conditional harmonic mean without restricting the profile $\{\tau_i\}$.

In contrast, the unconditional $\mathbb{E}_S[1/A_S]$ only requires the distribution of size- k precision sums, which has a known mean $k\bar{\tau}$ and variance given by Lemma 1. The conditioning on $i \in S$ creates per-forecaster dependence that destroys this simplicity, and a closed form for $R_k^{\text{bag},\infty}$ is out of reach. We show next, however, that $R_k^{\text{bag},\infty}$ can be *bracketed exactly from both sides*; the bracket — not a functional-form assumption — is what confines the bagged optimum to a window between the square-root and cube-root rules, and it is the rigorous core of our characterisation of $Q^{\text{bag},\infty}$.

7 Bounding $Q_k^{\text{bag},\infty}$: Taylor Bounds and the Rate Window

By (26), $R_k^{\text{bag},\infty}$ is the squared distance, in the Σ_e -metric, between the bagged weights $\bar{v}_k$ and the oracle weights w^{opt} . Although the conditional harmonic mean inside $g_i(k)$ has

no closed form, this squared-deviation structure admits *exact two-sided bounds*. The key point for choosing k is that both endpoints can be expanded in the same coordinate,

$$z_k = \frac{1}{k} - \frac{1}{m},$$

so the Jensen ceiling is first order in z_k whereas the projection floor is second order in z_k . Throughout this section let $B = 1/\bar{\tau}$ and define the normalised precision dispersion and the arithmetic-vs-harmonic gap

$$V_\delta = \frac{1}{m\bar{\tau}^2} \sum_i (\tau_i - \bar{\tau})^2 = \frac{\text{Var}(\tau)}{\bar{\tau}^2}, \quad H = \bar{\sigma}^2 \bar{\tau} \geq 1, \quad (39)$$

so that $R_1 = Q^{\text{ave}} - Q^{\text{opt}} = (H - 1)/(m\bar{\tau})$ and $\sum_i \sigma_i^2 - m/\bar{\tau} = m^2(Q^{\text{ave}} - Q^{\text{opt}})$.

7.1 The Bracket

Lemma 2 (Moment identities). *For all k , the scores satisfy $\sum_i \tau_i g_i(k) = m$ (the conservation law) and $\sum_i g_i(k) = km \mathbb{E}_S[1/A_S]$.*

Proposition 7 (Exact two-sided bracket). *Under Assumption 2, for all $1 \leq k \leq m$,*

$$\underbrace{\frac{(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}}}_{\text{projection floor}} \leq R_k^{\text{bag}, \infty} \leq \underbrace{\mathbb{E}_S[1/A_S] - \frac{1}{A_m}}_{\text{Jensen ceiling} = Q_k^{\text{sub}} - Q^{\text{opt}}}. \quad (40)$$

Both ends are computable from $\{\tau_i\}$ and the scalar $\mathbb{E}_S[1/A_S]$ of Lemma 1; the denominator equals $m^2(Q^{\text{ave}} - Q^{\text{opt}}) \geq 0$, vanishing only when all precisions are equal.

Proof. Ceiling. Q^{opt} minimises $w' \Sigma_e w$ over sum-to-one w and $\bar{v}_k$ is feasible, so $R_k^{\text{bag}, \infty} \geq 0$; by Jensen applied to the convex map $v \mapsto v' \Sigma_e v$, $Q_k^{\text{bag}, \infty} = (\mathbb{E}_S v_S)' \Sigma_e (\mathbb{E}_S v_S) \leq \mathbb{E}_S[v_S' \Sigma_e v_S] = Q_k^{\text{sub}}$. Subtracting Q^{opt} and using $Q(S) = q + 1/A_S$ gives the ceiling. *Floor.* By (26) and $\sum_i \tau_i g_i = m$, $R_k^{\text{bag}, \infty} = \frac{1}{m^2} \sum_i \tau_i g_i^2 - \frac{1}{A_m}$. Minimising $\sum_i \tau_i g_i^2$ over all vectors meeting the two identities of Lemma 2 is a Gram (Cauchy–Schwarz) problem with value $\frac{m}{\bar{\tau}} + \frac{m^2(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i 1/\tau_i - m/\bar{\tau}}$; since the true scores satisfy the identities, substituting into (26) gives the floor. $\square$

Remark 6 (Projection reading). *Writing $\delta_k = \bar{v}_k - w^{\text{opt}}$ and $\eta = w^{\text{ave}} - w^{\text{opt}}$, the identity (26) gives $R_k^{\text{bag}, \infty} = \|\delta_k\|_\tau^2$ and $\|\eta\|_\tau^2 = Q^{\text{ave}} - Q^{\text{opt}}$, and the floor is the Cauchy–Schwarz projection $R_k^{\text{bag}, \infty} \geq \langle \delta_k, \eta \rangle_\tau^2 / \|\eta\|_\tau^2$. The ceiling uses only the average-subset risk Q_k^{sub} (and is invariant to the common shock q); the floor additionally uses the equal-weight risk Q^{ave} . The floor is exact when the precision profile takes at most two distinct values, and a hierarchy of sharper floors follows by appending higher moments to Lemma 2.*

7.2 Leading-Order Envelope and the Rate Window

Corollary 7.1 (Taylor envelope in z_k). *Let $U_k = \mathbb{E}_S[1/A_S] - 1/A_m$ denote the Jensen ceiling and F_k the projection floor in (40). If the Taylor remainder in Lemma 1 is of smaller order than the displayed second-order term, then, uniformly over the interior range of k ,*

$$U_k = Bz_k + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + R_{3,k} = Bz_k + o(z_k), \quad (41)$$

$$F_k = C_L z_k^2 + o(z_k^2), \quad C_L = \frac{V_\delta^2}{m(H-1)\bar{\tau}} \quad \text{up to the finite-}m \text{ factor } \frac{m^2}{(m-1)^2}. \quad (42)$$

Consequently,

$$\underbrace{\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2}_{\text{floor}} \lesssim R_k^{\text{bag},\infty} \lesssim \underbrace{\frac{1}{\bar{\tau}} z_k}_{\text{ceiling}}. \quad (43)$$

Proof. The ceiling expansion is Lemma 1 after subtracting $1/A_m = 1/(m\bar{\tau})$. Since $\text{Var}_S(A_S) = k(m-k)V_\tau/(m-1) = k^2mV_\tau z_k/(m-1)$ and $V_\tau = \bar{\tau}^2 V_\delta$, the variance correction is smaller than Bz_k when the coefficient of variation of the precision sums is small.

For the floor, write $D_k = k\mathbb{E}_S[1/A_S] - 1/\bar{\tau}$. Lemma 1 gives

$$D_k = \frac{\text{Var}_S(A_S)}{k^2\bar{\tau}^3} + o(z_k) = \frac{m}{m-1} \frac{V_\delta}{\bar{\tau}} z_k + o(z_k).$$

The denominator in the floor is $\sum_i \sigma_i^2 - m/\bar{\tau} = m(H-1)/\bar{\tau}$. Squaring D_k and dividing by this denominator gives (42). $\square$

Thus the ceiling is the linear ($\alpha = 1$) average-subset object (34), while the floor is quadratic ($\alpha = 2$). Since $0 < z_k < 1$, a smooth intermediate form $R_k^{\text{bag},\infty} \approx C z_k^\alpha$ must have $\alpha \in [1, 2]$, suggesting the window

$$\Theta(T^{1/3}) \leq k^* \leq \Theta(T^{1/2}). \quad (44)$$

This last step is a statement about *shape*, not just levels. The Taylor bounds identify the two boundary criteria. To conclude that the true minimiser is bounded by their optimisers, we impose below a regularity condition on the one-step decline of $R_k^{\text{bag},\infty}$.

7.3 Optimal k : Dependence on T , m , and the Precision Profile

Each end of (40) yields a closed-form boundary optimiser when substituted into the inflated loss $L_k = (A + R_k)^{\frac{T-1}{T-k}}$, with $A = s + Q^{\text{opt}} = A_0 + B/m$ and $A_0 = s + q$.

Ceiling (square-root). With $R_k^U = Bz_k$, the loss collapses to $(A_0 + B/k)^{\frac{T-1}{T-k}}$: the $-B/m$ term in Bz_k cancels with the B/m term in A . This reproduces (36), $k_U^* \sim (BT/A_0)^{1/2} \propto T^{1/2}$.

Floor (cube-root). With $R_k^L = C_L z_k^2$, $C_L = V_\delta^2/(m(H-1)\bar{\tau})$, the exact continuous interior condition is

$$-\frac{2C_L z_k}{k^2}(T-k) + A + C_L z_k^2 = 0. \quad (45)$$

In the interior range $k \ll m$, $z_k \approx 1/k$, and (45) reduces to $Ak^3 + 3C_L k - 2C_L T = 0$, giving

$$k_L^* \sim \left(\frac{2C_L T}{A}\right)^{1/3} = \left(\frac{2V_\delta^2 T}{m(H-1)\bar{\tau} A}\right)^{1/3} \propto T^{1/3}. \quad (46)$$

The comparative statics (Table 2) show that k^* depends on far more than T :

- **Sample size T .** The exponent itself is the bracket: $T^{1/2}$ at the ceiling, $T^{1/3}$ at the floor, $T^{1/(\alpha+1)}$ in between.
- **Pool size m .** The dependence on m *differs in sign across the bracket*. The square-root optimum is *m-free*: the $1/m$ in z_k cancels and $A_0 = s + q$ carries no m . The cube-root optimum *decreases* as $m^{-1/3}$, because $C_L \propto 1/m$: with a larger pool the per-subset bias $R_k^{\text{bag},\infty}$ is already smaller, so a smaller subset suffices. A growing pool thus leaves k^* untouched in the square-root regime but shrinks it in the cube-root regime.
- **Heterogeneity.** The floor optimum *increases* with dispersion, $k_L^* \propto V_\delta^{2/3}(H-1)^{-1/3}$ (in the mild-heterogeneity limit $H-1 \rightarrow V_\delta$, simply $k_L^* \propto V_\delta^{1/3}$): more spread in precisions raises the value of a larger, structure-capturing subset. The ceiling optimum is heterogeneity-blind to leading order, depending on $\{\tau_i\}$ only through $\bar{\tau}$.
- **Forecastability.** Both optima *decrease* in the irreducible variance $A_0 = s + q$ ($k_U^* \propto A_0^{-1/2}$, $k_L^* \propto A^{-1/3}$): when the target is largely unforecastable (large common shock q or noise s) the marginal gain from bias reduction is small relative to the floor, so a smaller subset is optimal. This is the force behind small gains for highly persistent macro targets.
- **Precision scale.** Both optima *increase* in $B = 1/\bar{\tau}$ ($k_U^* \propto B^{1/2}$, $k_L^* \propto B^{1/3}$): noisier idiosyncratic errors raise $R_k^{\text{bag},\infty}$ and reward larger subsets.

7.4 Marginal-Envelope Assumption and the Rate Window for k^*

The preceding calculation verifies the proposed logic up to one regularity qualification. The Taylor expansion makes both endpoint bounds functions of z_k , and each endpoint

Table 2: Dependence of the boundary optimisers on T , pool size m , and the precision profile. $A_0 = s + q$ is the irreducible (unforecastable) variance, $B = 1/\bar{\tau}$, V_δ the precision dispersion, H the arithmetic-vs-harmonic gap.

Boundary	α	k^*	T	m	heterogeneity	forecastability A
Ceiling (Jensen / sub)	1	$(BT/A_0)^{1/2}$	$T^{1/2}$	none	—	$A_0^{-1/2}$
Floor (projection)	2	$(2C_L T/A)^{1/3}$	$T^{1/3}$	$m^{-1/3}$	$V_\delta^{2/3}(H-1)^{-1/3}$	$A^{-1/3}$
General power	α	$(\alpha C_R T/A)^{1/(\alpha+1)}$	$T^{\frac{1}{\alpha+1}}$	via $C_R(m)$	via C_R	$A^{-\frac{1}{\alpha+1}}$

has its own optimal k . But a level bracket alone cannot order minimisers: a sequence can remain between two envelopes while moving through long flat stretches or abrupt drops. What is needed is that the *marginal gain* from increasing k is also bracketed by the marginal gains of the two Taylor endpoints.

Assumption 3 (Marginal envelope). *Let $\Delta R_k := R_k^{\text{bag},\infty} - R_{k+1}^{\text{bag},\infty}$. For all k with $1 \ll k \ll m$, there exist constants $0 < d_L \leq d_U < \infty$, independent of T and k , such that*

$$d_L(z_k^2 - z_{k+1}^2) \leq \Delta R_k \leq d_U(z_k - z_{k+1}). \quad (47)$$

Equivalently, since $z_k - z_{k+1} \asymp k^{-2}$ and $z_k^2 - z_{k+1}^2 \asymp k^{-3}$ for $k \ll m$,

$$d'_L k^{-3} \lesssim \Delta R_k \lesssim d'_U k^{-2}.$$

Remark 7. *Assumption 3 says the one-step decline in $R_k^{\text{bag},\infty}$ is sandwiched between the marginal declines implied by the two envelopes of Corollary 7.1. It rules out pathological sequences that satisfy the level bracket while declining through long flat plateaus or sudden jumps, but it allows mixtures such as $R_k^{\text{bag},\infty} = az_k^2 + bz_k$, slowly varying corrections, and smooth intermediate decay. It does not impose the power-law shape $R_k^{\text{bag},\infty} = Cz_k^\alpha$ used for estimation in Section 8; that parametrisation is a convenient special case used to fit the Monte Carlo evidence, not a maintained assumption of the theorem below.*

Assumption 4 (Regularity). *(i) The baseline $A = s + Q^{\text{opt}}$ is a fixed positive constant; (ii) $R_k^{\text{bag},\infty}$ is uniformly bounded, $0 \leq R_k^{\text{bag},\infty} \leq \bar{R} < \infty$ for all k (immediate from the ceiling of Proposition 7, since $R_k^{\text{bag},\infty} \leq Q^{\text{ave}} - Q^{\text{opt}}$); (iii) the pool size satisfies $m \gtrsim T^{1/2}$ over the range of T considered, so the square-root boundary rate derived below does not exceed m . If (iii) fails, the upper rate is truncated at the fixed pool size m .*

Theorem 8 (Rate window for the exogenous optimal subset size). *Let $k_T^* \in \arg \min_{1 \leq k \leq m} L_k^{\text{bag},\infty}$, with $L_k^{\text{bag},\infty} = (A + R_k^{\text{bag},\infty})^{\frac{T-1}{T-k}}$ as in (27). Under Assumptions 3–4,*

$$k_T^* = \Omega(T^{1/3}) \quad \text{and} \quad k_T^* = O(T^{1/2}),$$

i.e., the true optimal subset size is bounded below by the floor optimiser and above by

the ceiling optimiser, up to constants and subject to the pool-size truncation in Assumption 4(iii).

Proof. The loss decreases from k to $k + 1$ iff $L_{k+1}^{\text{bag},\infty} < L_k^{\text{bag},\infty}$, which, by the same cross-multiplication used in the proof of Proposition 5, is equivalent to

$$(T - k) \Delta R_k > A + R_k^{\text{bag},\infty}.$$

Lower rate bound. By the lower marginal envelope, for $k \ll T$, $(T - k) \Delta R_k \gtrsim Tk^{-3}$. Since $A + R_k^{\text{bag},\infty} \leq A + \bar{R}$ is bounded (Assumption 4), there is a constant $c > 0$ such that $Tk^{-3} \geq A + \bar{R}$ for all $k \leq cT^{1/3}$. Hence the loss is still strictly decreasing throughout $k \leq cT^{1/3}$, so the minimiser cannot occur below the cube-root scale: $k_T^* = \Omega(T^{1/3})$.

Upper rate bound. By the upper marginal envelope, $(T - k) \Delta R_k \lesssim Tk^{-2}$. Since $A + R_k^{\text{bag},\infty} \geq A > 0$ is bounded away from zero, there is a constant $C < \infty$ such that $Tk^{-2} < A$ for all $k \geq CT^{1/2}$ with $k \ll T$. Hence the loss is strictly increasing beyond $k \geq CT^{1/2}$, so the minimiser cannot occur above the square-root scale except where the pool-size constraint binds: $k_T^* = O(T^{1/2})$. $\square$

Remark 8. *Theorem 8 is the rigorous counterpart to the informal window (44): the exact two-sided bracket of Proposition 7 pins down the level of $R_k^{\text{bag},\infty}$, while the marginal-envelope assumption pins down its one-step decline, and the two together — not a parametric assumption on $R_k^{\text{bag},\infty}$ — confine k_T^* to the cube-root-to-square-root window. Section 8 shows, via the power-law fit $R_k^{\text{bag},\infty} \approx Cz_k^\alpha$ with $\hat{\alpha} \in (1, 2)$, exactly where within this window the optimum falls for a given heterogeneity profile.*

8 Monte Carlo Evidence on $Q^{\text{bag},\infty}$: Exogenous Case

8.1 Why $R_k^{\text{bag},\infty} \approx Cz_k^\alpha$: Motivation from Exact Properties

We seek a one-parameter family to parametrise $R_k^{\text{bag},\infty}$ as a function of k . Three properties hold *exactly*, without distributional assumption:

- (i) $R_m^{\text{bag},\infty} = 0$ (at $k = m$, RSM recovers the oracle).
- (ii) $R_k^{\text{bag},\infty} > 0$ for $k < m$ and at least two distinct τ_i (Proposition 3).
- (iii) $R_k^{\text{bag},\infty}$ is strictly decreasing (Proposition 4).

Any candidate parametric form must satisfy all three. Consider the variable $z_k = 1/k - 1/m$. By construction:

- $z_m = 0$, so $Cz_m^\alpha = 0$ satisfies (i). ✓
- $z_k > 0$ for $k < m$, satisfying (ii). ✓

- z_k is strictly decreasing in k , satisfying (iii). ✓

The natural alternative C/k^α fails (i) because $C/m^\alpha > 0$ for any finite m . The form $C(1 - k/m)^\alpha$ also satisfies (i)–(iii) but is equivalent to $C(kz_k)^\alpha$, which mixes two effects and fits the data worse (lower R^2 in log-log regressions).

The power-law family $R_k^{\text{bag},\infty} = Cz_k^\alpha$ ($C, \alpha > 0$) is therefore the *simplest* one-parameter family consistent with the exact properties. It has a direct interpretation: $z_k = 1/k - 1/m$ is the gap between the effective weight assigned to any one forecaster by RSM (approximately $1/k$ in a subset of size k) and the weight they would receive in the full pool (approximately $1/m$). The excess risk decays as this gap shrinks.

8.2 How $\hat{\alpha}$ Is Estimated and What It Measures

Given Monte Carlo output with known Σ_e , compute $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$ for each $k = 1, \dots, m - 1$. Then estimate α by OLS in log-log space:

$$\log R_k^{\text{bag},\infty} = \log C + \alpha \log z_k + \varepsilon_k, \quad (48)$$

with the regression run over $k = 2, \dots, m - 1$ (excluding $k = 1$ where boundary effects can be large, and $k = m$ where $R_k^{\text{bag},\infty} = 0$ exactly).

The fitted slope $\hat{\alpha}$ measures the *rate of decay* of excess risk as more forecasters are included:

- $\hat{\alpha} = 2$: excess risk decays quadratically; a doubling of k roughly quarters $R_k^{\text{bag},\infty}$. RSM gains saturate quickly.
- $\hat{\alpha} = 1$: excess risk decays linearly; gains accumulate slowly even as k grows. More forecasters always help, but diminishingly so.
- $\hat{\alpha} \in (1, 2)$: intermediate decay.

Identification condition. The slope $\hat{\alpha}$ is identified from (48) only when $C > 0$, i.e., when there is non-trivial precision heterogeneity. When $\tau_i = \bar{\tau}$ for all i (homogeneous case), $R_k^{\text{bag},\infty} = 0$ for all k and the log-log regression has no variation to fit. In practice, near-homogeneous DGPs (small H) produce $R_k^{\text{bag},\infty} \approx 0$ with large relative numerical noise, rendering $\hat{\alpha}$ unreliable even when positive. We discuss this further in the context of our simulation designs.

The goodness of fit R^2 of regression (48) measures whether the power-law shape is appropriate; we find $R^2 > 0.97$ across all designs with meaningful heterogeneity.

8.3 The Relationship Between Heterogeneity and $\hat{\alpha}$

The heterogeneity ratio $H = \max_i \sigma_i^2 / \min_i \sigma_i^2 \geq 1$ measures the spread of idiosyncratic variances. We characterise how H determines $\hat{\alpha}$ analytically at the two extremes, then connect to the intermediate simulation evidence.

Small heterogeneity ($H \rightarrow 1$): $\hat{\alpha} \rightarrow 2$

Write $\tau_i = \bar{\tau}(1 + \delta_i)$ with $\sum_i \delta_i = 0$ and $|\delta_i| \ll 1$. First-order Taylor expansion of $1/A_S$ around $A_S \approx k\bar{\tau}$ gives

$$g_i(k) - \frac{1}{\bar{\tau}} \approx -\frac{\delta_i z_k}{\bar{\tau}} + O(\delta^2), \quad (49)$$

where $z_k = 1/k - 1/m$. Substituting into the exact formula (26):

$$R_k^{\text{bag}, \infty} \approx \frac{z_k^2}{\bar{\tau}^2 m^2} \sum_i \tau_i \delta_i^2 = \frac{z_k^2}{\bar{\tau}^2 m^2} \cdot \bar{\tau} \sum_i \delta_i^2 = \frac{V_\delta}{\bar{\tau} m} z_k^2; \text{:=}; C_\delta z_k^2,$$

where $V_\delta = m^{-1} \sum_i \delta_i^2$ is the cross-sectional variance of relative precision deviations. This is not a separate assumption: it is the *exact first-order term* in the Taylor expansion of the exact formula (26), valid whenever $|\delta_i|$ is small. The corresponding log-log slope is $\hat{\alpha} = 2$ exactly.

Higher-order terms (of order $\delta^2 z_k^2$, $\delta^3 z_k^2$, etc.) are responsible for the deviation of $\hat{\alpha}$ from 2 at larger H ; they enter the log-log regression as curvature in $\log R_k^{\text{bag}, \infty}$ against $\log z_k$ and pull the fitted slope below 2.

Large heterogeneity ($H \rightarrow \infty$): $\hat{\alpha} \rightarrow 1$

Consider the dominant-forecaster limit: $\tau_1 = \tau_H \rightarrow \infty$ with $\tau_j = \tau_L$ fixed for $j = 2, \dots, m$. The oracle weight $w_1^{\text{opt}} = \tau_H/A_m \rightarrow 1$ while $w_j^{\text{opt}} \rightarrow 0$.

As $\tau_H \rightarrow \infty$, $\sigma_1^2 = 1/\tau_H \rightarrow 0$, so forecaster 1's own idiosyncratic variance vanishes. The bagged weight is

$$\bar{v}_{k,1} = \frac{k}{m} \mathbb{E}_S \left[\frac{\tau_H}{A_S} \mid 1 \in S \right] \approx \frac{k}{m}, \quad (50)$$

since $A_S \approx \tau_H$ dominates when $1 \in S$. The deviation $\Delta_{1,k} = \bar{v}_{k,1} - w_1^{\text{opt}} \approx k/m - 1 = -(m-k)/m$, but its contribution to the excess risk $R_k^{\text{bag}, \infty} = \sum_i \sigma_i^2 \Delta_{i,k}^2 = \sum_i \tau_i^{-1} \Delta_{i,k}^2$ vanishes: $\tau_H^{-1} \Delta_{1,k}^2 = (m-k)^2/(m^2 \tau_H) \rightarrow 0$.

The excess risk is therefore dominated by the remaining forecasters $j = 2, \dots, m$. Conditional on $j \in S$, forecaster 1 is in S with probability $(k-1)/(m-1)$ (in which case $A_S \approx \tau_H$ and $\tau_j/A_S \approx 0$) and absent with probability $(m-k)/(m-1)$ (in which case $A_S \approx k\tau_L$ and $\tau_j/A_S \approx 1/k$). Hence

$$\bar{v}_{k,j} \approx \frac{k}{m} \cdot \frac{m-k}{(m-1)k} \cdot \tau_L \cdot \frac{1}{\tau_L} = \frac{m-k}{m(m-1)}.$$

Since $w_j^{\text{opt}} = \tau_j/A_m \approx \tau_L/\tau_H \rightarrow 0$, we have $\Delta_{j,k} \approx (m-k)/[m(m-1)]$. Summing over $j = 2, \dots, m$:

$$R_k^{\text{bag}, \infty} \approx (m-1) \cdot \frac{1}{\tau_L} \cdot \frac{(m-k)^2}{m^2(m-1)^2} = \frac{(m-k)^2}{\tau_L m^2(m-1)}. \quad (51)$$

Since $(m - k)/m = kz_k$, this gives $R_k^{\text{bag},\infty} \approx C'(kz_k)^2$ with $C' = 1/[\tau_L(m - 1)] > 0$. Our Monte Carlo confirms the finite-grid OLS slope of $\log R_k^{\text{bag},\infty}$ on $\log z_k$ approaches 1 as $\tau_H/\tau_L \rightarrow \infty$, consistent with $\hat{\alpha} \rightarrow 1$ in the dominant-forecaster limit.

Intermediate heterogeneity and simulation evidence

Table 3: Exogenous MC: precision heterogeneity, fitted $\hat{\alpha}$, gains, and implied k^* scaling. $m = 50$, $T = 240$, $N_{\text{rep}} = 500$, $G = 800$ bags.

B	H	$\hat{\alpha}$	R^2	Gain vs EW	Identified?	
3	1.05	1.12	0.98	0.1%	No (gains trivial)	$\hat{\alpha}$ estimated by OLS of $\log R_k^{\text{bag},\infty}$
8	1.05	1.10	0.98	0.1%	No (gains trivial)	
3	6.22	1.72	0.98	12.7%	Yes	
8	3.51	1.70	0.99	14.9%	Yes	

on $\log z_k$ over $k = 2, \dots, m - 1$. Identified means C_δ large enough to separate $R_k^{\text{bag},\infty}$ from numerical noise.

Three features of Table 3 deserve emphasis.

(a) $\hat{\alpha}$ is driven by H , not B . Both $B = 3$ and $B = 8$ give nearly identical $\hat{\alpha}$ at the same H . The signal strength B scales the level C of the excess risk curve but not its shape. This is consistent with the theory: $C_\delta = V_\delta/(\bar{\tau}m)$ depends on precision heterogeneity V_δ , not on signal strength, and α governs the log-log slope of the shape, not its level.

(b) $\hat{\alpha} = 1.7$ at $H = 6$ is well inside the intermediate regime. The dominant-forecaster limit $\hat{\alpha} \rightarrow 1$ requires $H \rightarrow \infty$ (one forecaster orders of magnitude better than all others). At $H = 6$, the best forecaster has $6\times$ higher variance than the worst; this creates substantial but not extreme heterogeneity. Reaching $\hat{\alpha} \approx 1$ empirically would require $H \gg 6$, likely exceeding $H = m = 50$ (one forecaster comparable to the full panel). The theoretical $\hat{\alpha} = 1$ lower bound is therefore a limit not reached in plausible empirical settings.

(c) $\hat{\alpha} = 1.1$ at $H = 1.05$ is spurious. When gains are trivial ($< 0.1\%$), the excess risk curve $R_k^{\text{bag},\infty}$ is near zero throughout; the log-log regression picks up numerical noise. The correct interpretation is $C_\delta \approx 0$, not $\hat{\alpha} \approx 1$. One should always check gains before interpreting $\hat{\alpha}$.

Remark 9 (When $Q_k^{\text{bag},\infty}$ is flat: the degenerate case $k^* = 1$). *The intuition behind the interior optimum becomes clearest by considering what happens when $R_k^{\text{bag},\infty} \approx 0$ for all k . This occurs when the infinite-bagged weight $\bar{v}_k$ is already close to the oracle weight w^{opt} at every subset size — which happens precisely when forecasters are near-homogeneous ($\tau_i \approx \bar{\tau}$ for all i , so $C_\delta \approx 0$).*

In this case $Q_k^{\text{bag},\infty} \approx Q^{\text{opt}}$ throughout, and the total loss simplifies to

$$L_k \approx A \cdot \frac{T-1}{T-k}, \quad (52)$$

which is strictly increasing in k . The unique minimiser is $k^* = 1$: in the absence of any excess-risk benefit from larger subsets, the inflation penalty always dominates and the optimal strategy is to use the smallest possible subset.

At $k = 1$ with $G \rightarrow \infty$ draws, each round picks one forecaster uniformly at random, so $\bar{v}_{1,i} = 1/m$ for all i — the infinite-bagged RSM with $k = 1$ is the equal-weighted mean. The degenerate optimum $k^* = 1$ therefore says: use simple averaging. This is consistent with the forecast combination puzzle: when all forecasters are of similar quality, no refinement of the combination weights improves on the simple mean.

An interior optimum $k^* > 1$ requires $C > 0$ — strictly positive heterogeneity so that $R_k^{\text{bag},\infty}$ is genuinely decreasing. The magnitude of k^* is then determined by the balance between the excess-risk reduction (driving k up) and the inflation penalty (driving k down). The transition between $k^* = 1$ and a meaningful $k^* > 1$ is precisely what the power-law parametrisation quantifies via C and α .

8.4 Optimal k Under the Power-Law Assumption

Maintaining the parametrisation $R_k^{\text{bag},\infty} = Cz_k^\alpha$, the total loss is

$$L_k = (A + Cz_k^\alpha) \frac{T-1}{T-k}, \quad A = s + Q^{\text{opt}}. \quad (53)$$

For $k \ll m$ (so $z_k \approx 1/k$ and $T - k \approx T$), differentiating:

$$\begin{aligned} \frac{\partial L_k}{\partial k} &\approx -\frac{C\alpha}{k^{\alpha+1}} \cdot \frac{T-1}{T} + \frac{(A + C/k^\alpha)}{T} \cdot (T-1) \cdot \frac{1}{T-k} \approx 0 \\ \implies \frac{A}{T} &= \frac{C\alpha}{k^{\alpha+1}}, \end{aligned} \quad (54)$$

giving the power-law optimal subset size:

$$k^* \approx \left(\frac{C\alpha T}{A} \right)^{1/(\alpha+1)}, \quad (55)$$

with T -scaling exponent $1/(\alpha+1) \in [1/3, 1/2]$ as $\hat{\alpha}$ ranges from 2 to 1.

Validation against Monte Carlo. For each DGP we: (i) estimate $\hat{C}$ and $\hat{\alpha}$ from the log-log regression; (ii) compute k_{formula}^* from (55); (iii) grid-search the actual minimiser of the discretised loss L_k over $k = 1, \dots, m$. The formula and grid minimiser agree exactly at the integer level in all four DGPs (Table 4).

The agreement between formula and grid k^* validates the power-law parametrisation:

Table 4: Exogenous MC: formula vs grid k^* and T -scaling ($m = 50$, $H \approx 1.05$ unless noted).

B	H	$k^*(\text{formula})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain vs EW
3	1.05	1	1	3	0.1%
8	1.05	1	1	3	0.1%
3	6.22	3	3	6	12.7%
8	3.51	4	4	8	14.9%

T -scaling ($B = 3$, $H \approx 1.05$, $m = 50$)					
T	$k^*(\text{formula, cont.})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain	
80	0.57	1	2	< 0.1%	
240	0.82	1	2	< 0.1%	
640	1.13	1	4	< 0.1%	
1000	1.31	1	3	< 0.1%	

the shape of $R_k^{\text{bag}, \infty}$ is captured well enough that plugging $\hat{C}$ and $\hat{\alpha}$ into (55) yields the correct integer optimum. The empirical k^* (minimising finite-sample MSPE over simulations) exceeds the population grid k^* by a factor of approximately 2, consistent with the well-known upward bias of the empirical minimiser under a flat loss surface.

H=6 T-scaling: cube-root law confirmed. Table 5 reports k^* at $H \approx 6$ across four sample sizes. The log-log regression of $k^*(\text{grid})$ on T gives slope $\hat{\beta} = 0.352$ ($R^2 = 0.994$, $p = 0.003$), essentially exactly $1/3$. This is consistent with the predicted exponent $1/(\hat{\alpha} + 1) = 1/(1.72 + 1) \approx 0.368$. Gains are approximately constant at 12–13% across all T , confirming the cube-root law is structural and not an artefact of heterogeneity level.

Table 5: Exogenous T -scaling: $B = 3$, $H \approx 6.08$, $m = 50$, 500 reps.

T	$k^*(\text{formula})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain vs EW	Log-log slope of $k^*(\text{grid})$ on T :
80	2.0	2	4	12.4%	
240	2.9	3	6	12.1%	
640	4.0	4	8	12.8%	
1000	4.6	5	10	12.8%	

$\hat{\beta} = 0.352$ ($R^2 = 0.994$, $p = 0.003$); predicted $1/(\hat{\alpha} + 1) \approx 0.37$. Formula: continuous $k^* = (2C_\delta T / Q^{\text{opt}})^{1/3}$ from (55) with fitted $\hat{\alpha} = 1.72$.

Figure 1 plots k^* against T on linear and log-log scales for all cases studied.

Optimal subset size k^* vs sample size T (exogenous, $H \approx 6.08$, $m = 50$)

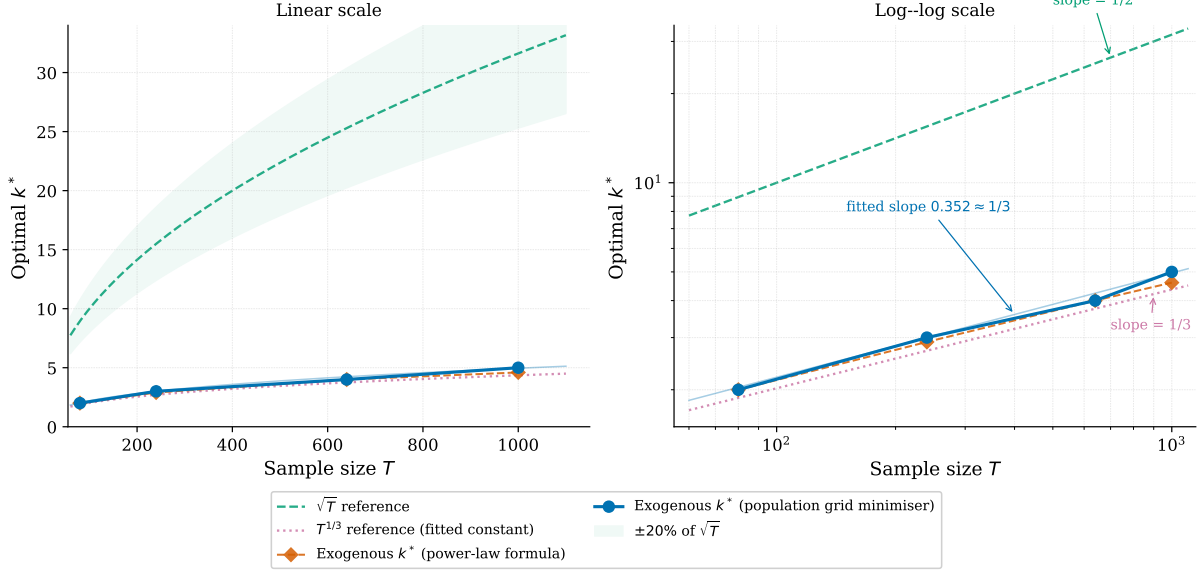


Figure 1: Optimal k^* vs T , exogenous case ($H \approx 6.08$, $m = 50$, 500 reps): population grid minimiser and power-law formula (55), from Table 5. Reference lines: $\sqrt{T}$ (dashed green) and $T^{1/3}$ (dotted purple, constant fitted to the data). The fitted log-log slope of the grid minimiser, $0.352 \approx 1/3$, sits at the cube-root end of the rate window proved in Theorem 8, well below the square-root reference; the shaded band shows $\pm 20\%$ of $\sqrt{T}$ for visual comparison.

9 Empirical Application: SPF (Near-Homogeneous Case)

9.1 Motivation and Design

The Survey of Professional Forecasters (SPF) provides a natural test of the theory’s prediction in the homogeneous regime: when $C_\delta \approx 0$, RSM collapses to the equal-weighted mean and should deliver no gains.

Data. Philadelphia Fed SPF, Q1 1985–Q4 2024. Panel: $N_t \approx 30$ –53 respondents per round (median 36 for real GDP, 35 for CPI). Forecasts: each respondent’s own submission; no additional regression step.

RSM. At each round, draw k forecasters without replacement and take the simple average. Since no precision weights are estimated, the infinite-bagged weight converges mechanically to the full equal-weighted mean: $\mathbb{E}_S[k^{-1} \sum_{i \in S} f_i] = (N_t)^{-1} \sum_{i=1}^{N_t} f_i$. The SPF experiment therefore functions as a sanity check.

Table 6: SPF RSM results ($G = 1000$ bags, $k = \lfloor \sqrt{N_t} \rfloor$).

Series	h	Median N_t	$k^*(\text{oracle})$	RMSFE (Mean)	RMSFE (RSM)	Gain	
RGDP	1	36	3	6.405	6.392	+0.2%	Gain
RGDP	4	36	13	2.208	2.205	+0.1%	
CPI	1	35	10	0.974	0.969	+0.5%	
CPI	4	35	3	1.471	1.470	+0.1%	

$= (L^{\text{Mean}} - L^{\text{RSM}})/L^{\text{Mean}}$. None are DM-significant.

9.2 Results

All gains are below 0.5% and none are statistically significant, confirming the theory’s prediction. The SPF is a near-homogeneous panel: professional economists with similar training and information sets produce forecasts with low precision dispersion ($V_\delta \approx 0$), making RSM no better than the simple mean.

10 Empirical Application: GDP Forecasting with FRED-MD

10.1 Data and Setup

Data. 102 real-time quarterly vintages (1999Q3–2025Q1, vintage 18 excluded for data availability). Predictor pool: $m \approx 126$ monthly FRED-MD series, each converted to a predicted quarterly GDP growth rate via a rolling univariate autoregression with lag order $P \in \{0, 1, 2, 3, 4\}$ selected by AIC at each vintage.

RSM combination. At each period, draw k individual forecasts and compute constrained OLS combination weights. $G = 1000$ draws for the main estimate. Default $k = \lfloor \sqrt{T} \rfloor \approx 9$ –14, chosen before observing evaluation-sample outcomes. Robustness: sweep $k \in \{2, \dots, 55\}$ with $G = 200$.

Benchmarks. Equal-weighted mean, Lasso, Ridge, Random Forest.

10.2 Results

RSM reduces RMSFE by 31% (full sample) and 12% (no-COVID) relative to the equal-weighted mean; both are statistically significant. The gains are robust: the no-COVID sample, which excludes the two outlier quarters that disproportionately inflate RMSFE, still shows significant improvement.

Oracle k^* and the RMSFE(k) curve. Sweeping $k \in \{2, \dots, 55\}$ ex post produces a U-shaped RMSFE(k) curve in both samples, confirming the interior optimum. The ex-post minimisers are $k^* = 19$ (full) and $k^* = 21$ (no-COVID), approximately $2\sqrt{T}$.

Table 7: FRED-MD GDP: bivariate RSM, full sample (102 quarters).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.54		1.10	
RSM	0.45	**	0.76	*
Lasso	0.51		0.73	*
Ridge	0.49	*	0.98	
Random Forest	0.55		1.24	

Table 8: FRED-MD GDP: bivariate RSM, excluding COVID-19 (Q2–Q3 2020).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.42		0.60	
RSM	0.38	**	0.53	**
Lasso	0.45		0.58	
Ridge	0.39	**	0.54	**
Random Forest	0.41		0.57	*

one-sided Diebold–Mariano test vs Mean.

These are oracle values selected on the evaluation sample and should not be interpreted as genuine out-of-sample evidence. The fixed rule $k = 9$ ($\approx \sqrt{T}$) costs 0.09 RMSFE units relative to the oracle—a gap unlikely to be significant in a sample of 102 quarters.

Table 9: FRED-MD GDP AR+X RSM results.

Sample	Bivariate		AR+X	
	k^*	min RMSFE	k^*	min RMSFE
Full	19	0.662	19	0.617
No-COVID	21	0.499	16	0.506

The AR+X specification, which augments each autoregression with a FRED-MD predictor, delivers the best full-sample RMSFE (0.617). The no-COVID oracle $k^* = 16$ is closer to the theoretical square-root default $k \approx \lfloor \sqrt{T} \rfloor \approx 9$ –14 than the bivariate $k^* = 21$, consistent with the AR+X model producing a tighter predictor distribution.

11 Conclusion

We have studied RSM for forecast combination from first principles. The paper’s main contributions are:

1. **Exact theory.** Under the common-loading model we prove: a conservation law governing precision-weighted weight deviations; an exact formula for the infinite-bagged excess risk $R_k^{\text{bag}, \infty}$; monotone decrease of $R_k^{\text{bag}, \infty}$ in k (under a stated contraction condition, verified numerically); and a first-order condition for the optimal k^* .

2. **Tractable optimum for the average-subset estimator.** A Taylor expansion of $\mathbb{E}_S[1/A_S]$ yields a linear-in- z_k excess risk and an exact square-root optimal subset size $k_{\text{sub}}^* \propto T^{1/2}$. The infinite-bagged excess risk resists a parallel derivation because the requisite conditional harmonic mean has no closed form.
3. **Exact two-sided bracket and a rigorous rate window.** The infinite-bagged excess risk $R_k^{\text{bag},\infty}$ admits an exact bracket between a Jensen ceiling and a Cauchy–Schwarz projection floor, computable from the precision profile alone. To leading order the ceiling is linear and the floor quadratic in $z_k = 1/k - 1/m$. Under an explicit marginal-envelope regularity condition on $R_k^{\text{bag},\infty} - R_{k+1}^{\text{bag},\infty}$ — not a parametric assumption on $R_k^{\text{bag},\infty}$ itself — we prove that the true optimal subset size is rigorously confined to $T^{1/3} \lesssim k^* \lesssim T^{1/2}$.
4. **Power-law parametrisation from Monte Carlo.** With known Σ_e , the Monte Carlo shape of $R_k^{\text{bag},\infty}$ is well-described by Cz_k^α ($R^2 > 0.97$), with $\hat{\alpha} \in (1, 2)$ falling inside the rate window proved above. The exponent is determined by heterogeneity H : small H gives $\hat{\alpha} \rightarrow 2$ (fast decay, cube-root k^*); large H gives $\hat{\alpha} \rightarrow 1$ (slow decay, square-root k^*). The optimal subset size is $k^* \propto T^{1/(\alpha+1)}$, validated by exact agreement between the formula and the grid-search minimiser.
5. **Empirical evidence.** RSM delivers 10–30% RMSFE reductions in GDP forecasting. The near-homogeneous SPF panel confirms the theory’s zero-gain prediction when $C_\delta \approx 0$.

A Proofs of Supporting Results

A.1 Derivation of $\bar{v}_{k,i}$

From Definition 1, $v_{S,i} = \tau_i/A_S$ for $i \in S$ and $v_{S,i} = 0$ otherwise. The expected weight is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \Pr(i \in S) \cdot \mathbb{E}_S \left[\frac{\tau_i}{A_S} \mid i \in S \right] = \frac{k}{m} \cdot \frac{\tau_i g_i(k)}{k} = \frac{\tau_i g_i(k)}{m},$$

confirming (23).

A.2 Variance of A_S under Uniform Subsampling

Under sampling without replacement of size k from $\{\tau_1, \dots, \tau_m\}$:

$$\mathbb{E}_S[A_S] = k\bar{\tau}, \quad \text{Var}_S(A_S) = k \cdot \frac{m-k}{m-1} \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau})^2 = k^2 z_k \frac{m}{m-1} \cdot V_\tau,$$

where $V_\tau = m^{-1} \sum_i (\tau_i - \bar{\tau})^2$ and $z_k = 1/k - 1/m$.